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BIPOLAR DISSECTOR

Abstract

A bipolar dissector includes a handle actuatable by squeezing, a housing extending from and coupled to a distal end of the handle, and a shaft coupled to a proximal end of the handle and extending past the distal end of the handle and through the housing to a distal end of the housing. The shaft includes first and second electrical lines extending from a proximal end to a distal end of the shaft and an insulating material electrically insulating the electrical lines from each other and from the handle and the housing. The bipolar dissector also includes a pair of forceps including a first tine and a second tine. The tines extend from the electrical lines at the distal end of the shaft at the distal end of the housing. Squeezing the handle actuates the forceps in the same direction as the squeezing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to and is a continuation of U.S. Nonprovisional application Ser. No. 18/345,734 filed on Jul. 6, 2023, which is a continuation of U.S. Nonprovisional application Ser. No. 16/566,419 filed Sep. 10, 2019 which issued as U.S. Pat. No. 11,717,338 on Aug. 8, 2023, which claimed the benefit of U.S. Provisional Application No. 62/729,306 filed Sep. 10, 2018, which are hereby incorporated by reference in their entirety.

FIELD

[0002] The present embodiments are directed to electrosurgical instruments. More particularly, the present embodiments are directed toward a bipolar dissector.

BACKGROUND

[0003] Electrosurgical forceps, typically referred to as bipolars or bipolar dissectors, provide the dual utility of mechanical manipulation of materials and tissue and application of a bipolar electrical current to provide for purposes of cutting, coagulating, dissecting, or fulgurating tissue. Such devices are well known in the art, and typically include either opposing tweezer-like extensions and a compressible tweezer-like handle or elongate extensions with tweezer-like or scissor-like pincers that are connected with and actuated by a pistol-type grip. The various commercial devices include elongate electrosurgical members that emit an electrical current at their opposing/cooperative distal tips. The electrical current is typically a radio-frequency alternating current. The electrosurgical members are controlled by a surgeon by actuation of the handle. The tweezer-type devices are generally lightweight and disposable. The pistol grip devices may be disposable or reusable.

[0004] A failing of current devices is the lack of a clamping grip action that can be actuated under the control of a tweezer-like action. Due to the space constraints and fine detail of many surgical procedures, use of an elongate pistol-activated device presents serious limitations for most surgeons, and introduces maneuverability, control, and precision problems. These problems may adversely affect the efficiency of the surgical procedure, creating undesirable delays and imprecision in the execution of the surgical manipulation of the tissue. There is a need for adaptations to such surgical instrumentation to provide precision and control together with functionalities that include one or more of suction and irrigation-type fluidic management in the device that integrates electrosurgical features.

BRIEF DESCRIPTION

[0005] In an embodiment, a bipolar dissector includes a handle actuatable by squeezing, a housing extending from and coupled to a distal end of the handle, and a shaft coupled to a proximal end of the handle and extending past the distal end of the handle and through the housing to a distal end of the housing. The shaft includes a first electrical line extending from a proximal end of the shaft to a distal end of the shaft, a second electrical line extending from a proximal end of the shaft to a distal end of the shaft, and an insulating material electrically insulating the first electrical line and the second electrical line from each other and from the handle and the housing. The bipolar dissector

also includes a pair of forceps including a first tine and a second tine. The first tine extends from the first electrical line at the distal end of the shaft at the distal end of the housing and the second tine extends from the second electrical line at the distal end of the shaft at the distal end of the housing. Squeezing the handle actuates the forceps in the same direction as the squeezing.

[0006] In some embodiments, the handle is offset from an axis of the housing.

[0007] In some embodiments, electrical leads are electrically coupled to the first electrical line and the second electrical line at the proximal end of the shaft and couplable to an electricity source.

[0008] In some embodiments, the shaft further includes a fluid channel extending from a proximal end of the shaft to the distal end of the shaft.

[0009] In some embodiments, the handle includes a pair of arms elastically squeezable to actuate the forceps.

[0010] In some embodiments, the distal end of the housing extends axially toward the forceps upon actuation of the handle and pushes on the forceps to cause distal tips of the forceps toward each other.

[0011] In some embodiments, the distal tips of the forceps are straight and splayed in a retracted state.

[0012] In some embodiments, the distal tips of the forceps are straight and parallel to each other in a retracted state.

[0013] In some embodiments, the first tine and the second tine each include a curved surface contacted by the housing upon actuation of the handle.

[0014] In some embodiments, the first tine and the second tine each include a flat surface contacted by the housing upon actuation of the handle.

[0015] In some embodiments, the housing is cylindrical.

[0016] In some embodiments, an axis of the housing is curved in a plane of the forceps.

[0017] In some embodiments, an axis of the housing is curved in a plane perpendicular to a plane of the handle.

[0018] In some embodiments, the housing is not straight in extending from the handle.

[0019] In another embodiment, a method of electrosurgery includes placing a pair of forceps of a bipolar dissector at an electrosurgical site. The bipolar dissector includes a handle, a housing extending from and coupled to a distal end of the handle, a shaft coupled to a proximal end of the handle and extending past the distal end of the handle and through the housing to a distal end of the housing, and the pair of forceps, including a first tine and a second tine. The shaft includes a first electrical line extending from a proximal end of the shaft to a distal end of the shaft, a second electrical line extending from a proximal end of the shaft to a distal end of the shaft, and an insulating material electrically insulating the first electrical line and the second electrical line from each other and from the handle and the housing. The first tine extends from the first electrical line at the distal end of the shaft at the distal end of the housing and the second tine extending from the second electrical line at the distal end of the shaft at the distal end of the housing. The method also includes squeezing the handle to actuate the forceps. Squeezing the handle actuates the forceps in the same direction as the squeezing.

[0020] In some embodiments, the method further includes applying suction to the electrosurgical site by way of a fluid channel through the shaft of the bipolar dissector.

[0021] In some embodiments, the method further includes providing irrigation to the electrosurgical site by way of a fluid channel through the shaft of the bipolar dissector.

[0022] In some embodiments, the squeezing further includes grabbing and pulling tissue at the electrosurgical site with the forceps.

[0023] In some embodiments, the method further includes supplying an electrical current to the forces by way of the first electrical line and the second electrical line.

[0024] Other features and advantages of the present invention will be apparent from the following

more detailed description, taken in conjunction with the accompanying drawings which illustrate, by way of example, the principles of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. **1** is a bottom view of a bipolar dissector in a retracted state in an embodiment of the present disclosure.

[0026] FIG. **2** is a side view of the bipolar dissector of FIG. **1**

[0027] FIG. **3** is an end view of the bipolar dissector of FIG. **1**.

[0028] FIG. **4** is an elevated perspective rear view of the bipolar dissector of FIG. **1**.

[0029] FIG. **5** is an elevated side view of the bipolar dissector of FIG. **1**.

[0030] FIG. **6** is a lowered side view of the bipolar dissector of FIG. **1**.

[0031] FIG. **7** is an elevated side view of a process of gripping with the bipolar dissector of FIG. **1**.

[0032] FIG. **8** shows forceps of a bipolar dissector with curved tines in an embodiment of the present disclosure.

[0033] FIG. **9** shows forceps of a bipolar dissector with flat tines in an embodiment of the present disclosure.

[0034] FIG. **10a** shows forceps of a bipolar dissector with tines having curved portions and flat portions and a parallel tip section in an embodiment of the present disclosure.

[0035] FIG. **10b** shows forceps of a bipolar dissector with tines having curved portions and flat portions and a splayed tip in an embodiment of the present disclosure.

[0036] FIG. **11** shows a side view of a bipolar dissector with an angled offset handle in an embodiment of the present disclosure.

[0037] FIG. **12** shows a lowered side view of a bipolar dissector with a non-straight housing in an embodiment of the present disclosure.

[0038] FIG. **13** shows a top view of a bipolar dissector with the housing curved in the plane of the forceps in an embodiment of the present disclosure.

[0039] FIG. **14** shows a lowered side view of a bipolar dissector with the housing curved in a plane perpendicular to the plane of the handle in an embodiment of the present disclosure.

[0040] Wherever possible, the same reference numbers will be used throughout the drawings to represent the same parts.

DETAILED DESCRIPTION

[0041] Provided is a surgical instrument for use in a wide variety of roles that may include, but are not limited to, grasping, dissecting, clamping, and/or retracting materials or tissue during surgical procedures performed in topical surgery, such as, for example, cosmetic surgery; in open surgery, such as, for example, brain surgery; or in surgery within the abdominal cavity.

[0042] Embodiments of the present disclosure, for example, in comparison to concepts failing to include one or more of the features disclosed herein, provide a squeezing force sufficient to grab and pull tissue in a bipolar instrument, provide actuation by squeezing of a handle in the same direction as the squeezing of the forceps in a bipolar instrument, provide irrigation in a bipolar instrument, provide suction in a bipolar instrument, and/or combinations thereof.

[0043] Referring to FIG. **1**, the bipolar dissector **10** includes a handle **12**, a shaft **14** coupled to and extending from the handle **12**, a pair of forceps **16** extending from the shaft **14** on the end opposite the handle **12**, and a housing **18** coupled to the handle **12** and extending around the shaft **14**. The pair of forceps **16** includes a first tine **16a** and a second tine **16b**. The proximal end **20** of the shaft **14** includes electrical leads **22** that are electrically connected with the forceps **16**. The center of the shaft **14** along its length lies within a plane perpendicular to the plane of the view of FIG. **1**. The bipolar dissector **10** is shown in a retracted state in FIG. **1** through FIG. **6**.

[0044] Referring to FIG. 2, the handle **12** of the bipolar dissector **10** is offset from the axis of the housing **18**. More specifically, the handle **12** lies in a plane substantially parallel to, but offset from, the plane in which the forceps **16** and the axis of the housing **18** lie. The shaft **14** also follows the offset of the handle **12** from the housing **18**, extending through both the handle **12** and the housing **18**. As shown in FIG. 1 and FIG. 2, the housing **12** is cylindrical in shape.

[0045] Referring to FIG. 3, the shaft **14** includes electrical lines **24** that separately conduct electricity from each of the two electrical leads **22** to each of the two forceps **16**. In some embodiments, each of the two forceps **16** is integral with its respective electrical line **24**. The shaft **14** also includes insulating material **26** that electrically insulates the electrical lines **24** from each other and from the handle **12** and the housing **18**. The shaft **14** may optionally contain a fluid channel **28** that fluidly connects the opposite ends of the shaft **14** and is also electrically insulated from the electrical lines **24**. The forceps **16** extend from the distal end **30** of the shaft **14**.

[0046] Referring to FIG. 4, the electrical leads **22** are each separately connectable to an appropriate electrical source (not shown), and the fluid channel **28** is connectable to tubing (not shown) at the proximal end **40** of the handle **12** for either providing suction to remove fluid from the surgical site or providing irrigation to apply fluid to the surgical site.

[0047] Referring to FIG. 5, FIG. 6, and FIG. 7, actuation of the handle **12** by squeezing the arms **50** of the handle toward the shaft **14**, such as by a hand of the user of the bipolar dissector **10**, causes the handle **12** to elastically deform. Being fixed to the shaft **14** at the proximal end **40**, the handle **12** elastically deforms to extend at the distal end **42** toward the forceps **16**, thereby pushing the housing **18** axially with respect to the shaft **14** toward the forceps **16**. The distal end **54** of the housing **18** extends to press on the outer surfaces of the angled proximal ends of the forceps **16**, causing the forceps **16** to come toward each other and eventually together at their distal tips in a pinching movement. FIG. 7 shows the bipolar dissector **10** in the retracted state and in the closed or unretracted state.

[0048] As shown in FIG. 7, the squeezing force is applied in the same direction as the direction of the closing of the forceps **16**. This gives the user a feeling of the squeezing force being connected to the closing action of the forceps **16**. The handle **12**, the housing **18**, and/or the forceps **16** may be biased toward the retracted state by one or more springs (not shown), such as, for example, leaf springs. The housing **18** may be integral with, or otherwise coupled to, the distal end **42** of the handle **12**. The housing **18** is not coupled to the shaft **14**. Removal of the squeezing force causes the handle **12**, the housing **18**, and the forceps **16** to elastically return to the retracted state.

[0049] In some embodiments, the surgical instrument includes a handle **12** formed of opposing compressible arms **50** that are attached at opposing ends to form a generally elongated elliptical shape, as shown in FIG. 5. In other embodiments, the shape of the handle **12** may be more rounded or may be squared. The instrument includes, attached to the handle **12**, a shaft **14** with a generally circular cross section that extends from proximal to distal ends of the instrument, passing through the handle **12** and extending beyond the distal end of the handle **12** to terminate at a distal end of the housing **18** for contact with and manipulation of tissue. The shaft **14** is covered by a solid housing **18** along at least a portion of its length and contains an extension member that terminates at the distal end in opposing grasping blades or tines **16a/16b** as a pair of forceps **16**. The cylindrical shaft **14** also contains one or more of suction and irrigation channels **28**, which may include separate channels for each, an electrical lead **22** for delivering electrical current at the forceps **16**, and an insulating material **26** to maintain separation between the irrigation channels **28** and the electrical lines **24**. Of course, in some embodiments, the shaft **14** may lack the irrigation port features and may thus include only insulating material **26** and electrical lines **24**.

[0050] In use, the handle **12** is grasped and squeezed to direct movement of the tubular housing **18** along a linear path in a direction from the proximal end **56** toward the distal end **54** so as to extend toward and over the tines **16a/16b** and thereby compress them. Squeezing the handle **12** actuates the forceps **16** in the same direction as the squeezing direction. Release of compression on the

handle **12** causes the housing **18** to retract towards the proximal end **56**, thereby exposing more of the tines **16a/16b** and effectively releasing them from compressed contact. Adjustable compression of the tines **16a/16b** enables the user to control the gripping and release of tissue between the forceps **16**. Controllers and actuators (not shown) connected to the irrigation channel **28** and electrical lines **24** enable controlled actuation of those features. The controller/actuator may be located remotely from the bipolar dissector **10**, such as, for example, in the form of one or more foot pedals actuated by a foot of the user to start or stop flow of electrical current, flow of irrigation fluid, and/or application of suction. The bipolar dissector **10** may be formed of various materials. In some embodiments, the handle **12** and tubular housing **18** may be formed of plastic, and at least the forceps **16** may be formed of metal at their distal ends and may be attached to opposing arms that are formed of metal or plastic. The shaft **14** within the housing **18** may be formed of plastic, and such plastic may include insulative material.

[0051] The bipolar dissector **10** may have any appropriate dimensions. In some embodiments, the housing **18** is about 15 cm in length. In some embodiments, a cross section of the cylindrical shaft **14** has a diameter of about 3.5 mm, although the shaft **14** may have any appropriate cross sectional shape, such as, for example, elliptical, oval, square, rectangular, or polyhedral. In some embodiments, the forceps have a length of about 15 mm and a thickness of about 5 mm.

[0052] In some embodiments, the forceps **16** are metal with no insulating cover. In other embodiments, at least a portion of the forceps **16**, starting from the proximal end, is covered with an electrical insulation, such as, for example, to prevent metal-metal contact between the housing **18** and the forceps **16** during actuation, but the distal tips are not. More generally, when any portion of the housing **18** is made of an electrically-conductive metal or another electrically-conductive material, the contact surfaces for contact between the housing **18** and the forceps **16** during actuation include one surface being non-electrically conductive to prevent electrical current from flowing into the housing **18**.

[0053] Referring to FIG. **8**, FIG. **9**, FIG. **10a**, and FIG. **10b**, the forceps **16** may have any appropriate shape complementary to the housing **18** such that axial extension of the housing **18** applies a force to surfaces of the forceps **16** sufficient to bring the distal tips of the tines **16a/16b** into contact with each other. Appropriate shapes include tines **16a/16b** that angle away from the axis of the housing **18** farther than the radial length of the housing **18** from the proximal end of the tines **16a/16b** toward a mid-portion of the tines **16a/16b** with the tines **16a/16b** angling toward each other and toward the axis of the housing **18** at or toward the distal tips.

[0054] Referring to FIG. **8**, a portion of the tines **16a/16b** may be curved to angle away from the axis of the housing **18** and provide the contact surface for contacting the housing **18** to bring the distal tips of the forceps **16** together and may be curved to angle back toward the axis of the housing **18** toward the distal tips. Referring to FIG. **9**, the tines **16a/16b** may include angled flat, straight sections to angle away from the axis of the housing **18** and provide the contact surface for contacting the housing **18** to bring the distal tips of the forceps **16** together and may include another angled flat section to angle back toward the axis of the housing **18** toward the distal tips.

Additionally or alternatively, the tines **16a/16b** may cross over each other between the proximal end and the distal end, as shown in FIG. **9**. Referring to FIG. **10a** and FIG. **10b**, a portion of the tines **16a/16b** may be angled and flat to provide the contact surface for contacting the housing **18** and another portion of the tines **16a/16b** may be curved to bring the distal tips of the forceps **16** together.

[0055] The distal portions of the tines **16a/16b** in FIG. **8**, FIG. **9**, FIG. **10a**, and FIG. **10b** are flat. The distal portions of the tines **16a/16b** in FIG. **8**, FIG. **9**, and FIG. **10a** are parallel in the retracted states and come together to contact at their distal tips for gripping and pulling upon actuation. The distal portions of the tines **16a/16b** in FIG. **10b** are splayed in the retracted state and come together substantially parallel to each other upon actuation to provide a greater area of contact between the tines **16a/16b** for gripping and pulling upon actuation.

[0056] Alternatively to the offset of the handle **12** shown in FIG. **1** to FIG. **7**, the handle **12** may be offset from the forceps **16** in other manners. Referring to FIG. **11**, the handle **12** may be in a plane that is offset and at an angle to the plane of the forceps **16** rather than being offset and in a plane parallel to the plane of the forceps **16**.

[0057] Alternatively to the housing **18** and the shaft **12** under the housing **18** being cylindrical as shown in FIG. **1** to FIG. **7**, the axis of the housing **18** may be curved or otherwise not straight. In such embodiments, the shaft **12** may be flexible under a rigid housing **18** to conform to the contour of the housing **18** during activation and retraction. Alternatively, the housing **18** may be flexible over a rigid shaft **12** to conform to the contour of the shaft **12** during activation and retraction. Referring to FIG. **12**, the housing **18** has an irregular contour in extending from the handle **12**. Referring to FIG. **13**, the housing **18** has a curved contour in extending from the housing **18**, where the curve is in the plane of the forceps **16**. Referring to FIG. **14**, the housing **18** has a curved contour in extending from the housing **18**, where the curve is in the plane perpendicular to the plane of the handle **12**.

[0058] Embodiments of the present invention are not in any way limited in use for any particular type of surgery, and thus, may be use in spinal surgery in any mode of access, including, but not limited to direct surgical approaches, and minimally invasive approaches including direct, oblique or posterior-lateral, anterior (ALIF), posterior (PLIF), transverse (TLIF), lateral, and extreme lateral (XLIF), as well as in any other surgeries in the body, including within the brain, the bone extremities, and in the pelvis and abdomen.

[0059] As used herein, the singular forms “a”, “an”, and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. The term “proximal” as used in connection with any object refers to the portion of the object that is closest to the operator of the object (or some other stated reference point), and the term “distal” refers to the portion of the object that is farthest from the operator of the object (or some other stated reference point). The term “operator” means and refers to any professional or paraprofessional who delivers clinical care to a medical patient, particularly in connection with the delivery of care.

[0060] With respect to any references herein that may be made relative to a human patient, the terms “cephalad”, “cranial”, and “superior” indicate a direction toward the head, and the terms “caudad” and “inferior” indicate a direction toward the feet. Likewise the terms “dorsal” and “posterior” indicate a direction toward the back, and the terms “ventral” and “anterior” indicate a direction toward the front. And the term “lateral” indicates a direction toward a side of the patient, the term “medial” indicates a direction toward the mid line of the patient, and away from the side, the term “ipsilateral” indicates a direction toward a side that is proximal to the operator or the object being referenced, and the term “contralateral” indicates a direction toward a side that is distal to the operator or the object being referenced. More generally, all terms providing spatial references to anatomical features shall have meaning that is customary in the art.

[0061] Unless otherwise indicated, all numbers expressing quantities, properties, and so forth as used in the specification, drawings and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless otherwise indicated, the numerical properties set forth in the specification and claims are approximations that may vary depending on the suitable properties desired in embodiments of the present invention. Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the general inventive concepts are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical values, however, inherently contain certain errors necessarily resulting from error found in their respective measurements.

[0062] References to visualization using radiography as described in the exemplary techniques herein are merely representative of the options for the operator to visualize the surgical field and the patient in one of many available modalities. It will be understood by one of ordinary skill in the art that alternate devices and alternate modalities of visualization may be employed depending on

the availability in the operating room, the preferences of the operator and other factors relating to exposure limits. While confirmation of instrument placement in the course of the technique is appropriate, the frequency and timing relative to the sequence of steps in the technique may be varied and the description herein is not intended to be limiting. Accordingly, more or fewer images, from more or fewer perspectives, may be collected.

[0063] One of ordinary skill will appreciate that references to positions in the body are merely representative for a particular surgical approach, and according to the exemplary embodiments herein, are suitable for any number of animal patients, including humans and other species. Of course, the type of surgery, target tissue, and species of patient may be different than is disclosed in the exemplary embodiments described herein. Further, all references herein are made in the context of the representative images shown in the drawings. Fewer or additional generic instruments may be used according to the preference of the operator. Moreover, references herein to specific instruments are not intended to be limiting in terms of the options for use of other instruments where generic options are available, or according to the preference of the operator.

[0064] Thus, while the disclosed embodiments have been described and depicted in the drawings in the context of the human spine, it should be understood by one of ordinary skill that all or various aspects of the embodiments hereof may be used in connection with other species and within any species on other parts of the body where deep access within the tissue is desirable.

[0065] Further, while various inventive aspects, concepts and features of the general inventive concepts are described and illustrated herein in the context of various exemplary embodiments, these various aspects, concepts and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the general inventive concepts. Still further, while various alternative embodiments as to the various aspects, concepts and features of the inventions (such as alternative materials, structures, configurations, methods, devices and components, alternatives as to form, fit and function, and so on) may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed.

[0066] Those skilled in the art may readily adopt one or more of the inventive aspects, concepts and features into additional embodiments and uses within the scope of the general inventive concepts, even if such embodiments are not expressly disclosed herein. Additionally, even though some features, concepts and aspects of the inventions may be described herein as being a preferred arrangement or method, such description is not intended to suggest that such feature is required or necessary unless expressly so stated. Still further, exemplary or representative values and ranges may be included to assist in understanding the present disclosure; however, such values and ranges are not to be construed in a limiting sense and are intended to be critical values or ranges only if so expressly stated.

[0067] Moreover, while various aspects, features and concepts may be expressly identified herein as being inventive or forming part of an invention, such identification is not intended to be exclusive, but rather there may be inventive aspects, concepts and features that are fully described herein without being expressly identified as such or as part of a specific invention. Descriptions of exemplary methods or processes are not limited to inclusion of all steps as being required in all cases, nor is the order that the steps are presented to be construed as required or necessary unless expressly so stated.

Claims

1. A bipolar dissector, comprising: a handle including a pair of elastically deformable arms that are configured to move towards each other; a housing extending from and coupled to a distal end of the handle; a shaft disposed within the handle and extends into the housing and through a distal end of

the housing, the shaft being co-axially aligned with the housing; and a pair of forceps disposed within the shaft, the pair of forceps includes a first tine and a second tine extending beyond the distal end of the housing; wherein the housing is configured to axially move with respect to the shaft such that the distal end of housing extends to contact the first and second tines, causing the first and second tines to move towards each other.

2. The bipolar dissector of claim 1, wherein the housing is configured to move along a linear path in a direction from a proximal end of the housing towards the distal end of the housing.

3. The bipolar dissector of claim 1, wherein the handle is offset from an axis of the housing.

4. The bipolar dissector of claim 3, wherein the shaft extends from a proximal end of the handle to a distal end of the handle and into the housing that is offset with respect to the handle.

5. The bipolar dissector of claim 1, wherein the handle is a generally elongated elliptical shape.

6. The bipolar dissector of claim 1, wherein the shaft further comprising: a first electrical line extending from a proximal end of the shaft to a distal end of the shaft; a second electrical line extending from a proximal end of the shaft to a distal end of the shaft; and an insulating material electrically insulating the first electrical line and the second electrical line from each other and from the handle and the housing.

7. The bipolar dissector of claim 1, wherein the shaft includes at least one suction channel and at least one irrigation channel.

8. The bipolar dissector of claim 1, wherein the housing is cylindrical.

9. The bipolar dissector of claim 1, wherein an axis of the housing is curved in a plane of the pair of forceps.

10. The bipolar dissector of claim 1, wherein an axis of the housing is curved in a plane perpendicular to a plane of the handle.

11. The bipolar dissector of claim 1, wherein the housing is non-planar in extending from the handle.

12. A bipolar dissector, comprising: a handle including a pair of elastically deformable arms that are configured to move towards each other; a housing extending from and coupled to a distal end of the handle; a shaft disposed within the handle and extends into the housing and through a distal end of the housing, the shaft being co-axially aligned with the housing; and a pair of forceps disposed within the shaft, the pair of forceps includes a first tine and a second tine extending beyond the distal end of the housing, wherein, in a first position of the handle, the housing detracts with respect to the shaft to cause a distal tip of the first and second tines to move towards each other, and wherein, in a second position of handle, the housing retracts with respect to the shaft to cause a distal tip of the first and second tines to move away from each other.

13. The bipolar bisector of claim 12, wherein the first position is when the pair of elastically deformable arms moves towards each other.

14. The bipolar bisector of claim 12, wherein the second position is when the pair of elastically deformable arms moves away from each other.

15. A method of electrosurgery comprising: placing a pair of forceps of a bipolar dissector at an electrosurgical site, the bipolar dissector including: a handle including a pair of elastically deformable arms that are configured to move towards each other; a housing extending from and coupled to a distal end of the handle; a shaft disposed within the handle and extends into the housing and through a distal end of the housing, the shaft being co-axially aligned with the housing; and a pair of forceps disposed within the shaft, the pair of forceps includes a first tine and a second tine extending beyond the distal end of the housing; and squeezing the handle to actuate the pair of forceps; wherein squeezing the handle, while coupled to the shaft at the proximal end, causes the housing to axially move with respect to the shaft such that the distal end of housing extends to contact the first and second tines, causing the first and second tines to move towards each other.

16. The method of claim 15, further comprising applying suction to the electrosurgical site by way

of a fluid channel through the shaft of the bipolar dissector.

17. The method of claim 15, further comprising providing irrigation to the electrosurgical site by way of a fluid channel through the shaft of the bipolar dissector.

18. The method of claim 15, wherein the squeezing further comprises grabbing and pulling tissue at the electrosurgical site with the pair of forceps.

19. The method of claim 15, further comprising supplying an electrical current to the forceps by way of the first electrical line and the second electrical line.

20. The bipolar dissector of claim 1, wherein the shaft is a tubular member that extends from the proximal end of the handle to the distal end of the handle and extends through the housing to the distal end of the housing.
